

AI Talent and Skills Market Analysis

Final Project Documentation — Python + Power BI

Project Objective:

The main objective of this project is to perform an end-to-end analysis of the AI job market for 2025–2026 using Python. The project covers data loading, cleaning, preprocessing, exploratory data analysis (EDA), statistical analysis, and visualization to identify meaningful patterns and trends in job roles, required skills, salaries, experience levels, job demand, locations, company size, and work type. The analysis aims to derive data-driven insights that can support better career and hiring decisions.

Dataset Information:

Item	Details
Dataset	AI Jobs Market 2025–2026
Size	1,500 rows × 25 columns
Domain	Artificial Intelligence / Technology / Employment
Source	Rasse23 GitHub Gist dataset
Tools	Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI
Main Areas	Roles, skills, salary, experience, demand, location, work type

Project Scope and Analysis Framework:

The scope of this project covers the complete analysis of the **AI job market for 2025–2026**, from data loading and cleaning to exploratory analysis, visualization, and insight generation.

1. Data Loading and Initial Overview :

- Imported the **AI Jobs Market 2025–2026** dataset using Pandas.
- Examined the dataset dimensions **1,500 rows and 25 columns**.
- Checked the data types of all columns using `info()`.
- Reviewed the initial records using `head()`.
- Generated statistical summaries using `describe()` to understand numerical variables. Examined column names and the overall dataset structure.

2. Data Pre-processing:

- Checked the dataset for missing values **no missing values were found**.
- Checked for duplicate records in the dataset.
- Identified **119 repeated skills within the Required Skills column and removed the repeated skills** while retaining the unique skills for each job posting.
- Removed the **Job_ID** column as it was only a unique identifier.
- Removed redundant columns such as **is_senior** and **is_remote_friendly**.
- Corrected and standardized column names and data types where required.
- Renamed **company_size** to Company Employee Range and **remote_work** to Work Type.
- Created derived columns such as Posting Period, Quarter, Salary Range USD, Monthly Salary USD, and Demand Growth YoY log.
- Prepared the cleaned dataset for further EDA and visualization.

3. Exploratory Data Analysis (EDA):

- Performed **univariate, bivariate, and multivariate analysis** to understand the AI job market.
- Analyzed job categories, company size, salary distribution, salary range, experience level, required skills, work type, industry, and job demand.
- Used **groupby and pivot-table analysis** to compare salary, skills, experience, demand, and other job characteristics.
- Performed **correlation analysis** to identify relationships between numerical variables such as salary, demand growth, salary premium, and experience.
- Used statistical summaries such as **mean, median, standard deviation, quartiles, minimum, and maximum values** to support the analysis.

4. Data Visualization:

Data visualization was performed using **Matplotlib and Seaborn** in Python to represent the patterns, trends, distributions, and relationships identified during Exploratory Data Analysis (EDA). Different visualization techniques were selected based on the type of variables and the purpose of the analysis.

Univariate Analysis

Univariate analysis was used to study individual variables separately.

Bar charts were used for categorical variables such as **Job Category, Company Size, Experience Level, Work Type, and Education Required** to understand the distribution and frequency of job postings.

Histograms were used for numerical variables such as **Annual Salary and Years of Experience** to understand their distribution and concentration.

Box plots were used for salary-related variables to identify the **median, spread, and possible outliers**.

Overall, the univariate visualizations helped identify the distribution of job postings, salary patterns, experience levels, company sizes, work types, and education requirements.

Bivariate Analysis

Bivariate analysis was used to study the relationship or comparison between two variables.

Bar charts were used to compare **average salary across different job roles and experience levels**, helping identify higher-paying roles and salary differences.

A **line chart** was used to show the **average salary trend across experience levels**. The result showed that **average salary generally increases as experience level increases**, with Lead-level roles having the highest average salary.

A **grouped bar chart** was used to compare **Work Type Distribution by Country** across On-site, Hybrid, and Fully Remote roles. The USA had the highest overall job-posting volume, while Hybrid work was prominent across many countries.

The relationship between **AI Salary Premium and Demand Growth** was also analyzed. The correlation value of **-0.085** indicated a very weak negative relationship between the two variables.

Multivariate Analysis

Multivariate analysis was used to study multiple variables together and identify detailed patterns in the AI job market.

A **heatmap** was used to analyze **Skills by Job Role**, helping identify the skills commonly required for different AI roles.

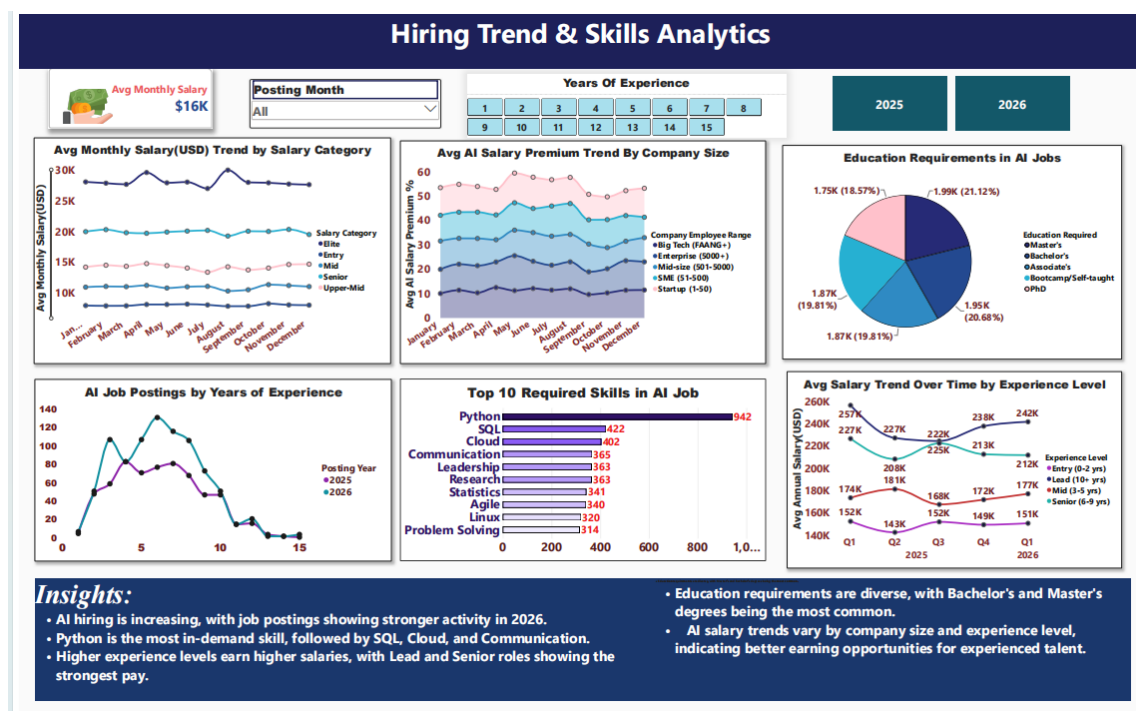
Additional comparative visualizations were used to examine multiple factors such as **Salary, Experience Level, Company Size, Industry, and Demand Score** together. These visualizations helped identify differences in salary and demand across different job-market segments.

Overall, the multivariate analysis provided a broader understanding of how **skills, job roles, experience, salary, company size, industry, and demand** are connected within the AI job market.

Overall Visualization Outcome

The Python visualizations converted the cleaned dataset into meaningful graphical outputs and helped identify **job distribution, salary trends, experience-based salary growth, work-type preferences, skill requirements, and demand patterns**. These findings were further used to develop the **Power BI dashboard and generate business-oriented insights and recommendations**.

Dashboard Screenshot:



5. Insight Generation and Report:

Key Insights:

- The AI job market shows strong hiring activity, with around 1,500 job postings across 14 countries.
- The USA has the highest number of AI job postings, indicating strong demand for AI professionals.
- Senior ML Engineer and LLM Engineer roles offer some of the highest average salaries.
- New York has the highest average annual salary among the analyzed cities.
- AI hiring activity increased in 2026, indicating growing demand for AI professionals.
- Python, SQL, Cloud, and communication skills are commonly required in AI job postings.
- Higher experience levels are generally associated with higher salaries.
- Bachelor's and Master's degrees are common education requirements for AI roles.
- AI salary premium varies across company sizes, suggesting differences in compensation.
- Overall, the AI job market shows strong growth potential, competitive salaries, and increasing demand for technical skills.

Overall Finding:

The analysis shows strong demand for specialized AI roles, with higher salaries generally associated with greater experience and advanced AI skills.

Conclusion:

Overall, the analysis shows that the AI job market is growing, with increasing hiring activity, competitive salaries, and strong demand for technical skills such as Python, SQL, and Cloud. Experienced professionals with relevant skills and qualifications can access better-paying AI opportunities.

Recommendations:

- Develop skills in **Python, SQL, Cloud, and AI/ML**.
- Gain practical experience through **projects and internships**.
- Build both **technical and communication skills**.
- Focus on advanced AI roles and relevant experience to improve career and salary opportunities.